# Lab E — Loudspeaker response in free field

Reference version (for checking)

34870 Electroacoustics

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**Reference version, not the hand-in.** This is a complete worked version of the lab, meant for checking the group's own report and figures: the numbers, the plots and the reasoning. Its figures sit in `LabE/report/reference_figures/`, apart from `LabE/figures/`, which is left for the group's own plots. Every number comes from the measured files `LabE/data/labE_<tag>.mat` and the dimensions in `LabE/matlab/labE_dims.m`, or is marked as an assumption. The group writes its own text.

## 1 Purpose and theoretical basis

The lab has two purposes: (a) to show how the frequency response and directivity of one loudspeaker unit depend on the baffle it is mounted in, and (b) to measure the magnitude and relative phase of the three units of the project loudspeaker (System D), which the crossover design needs.

### 1.1 Edge diffraction

A driver in an infinite baffle radiates into half space ($2\pi$). On a finite baffle the wave that travels along the front reaches the edge, and every edge point re-radiates a secondary wave of opposite sign. At the microphone the direct and the edge waves add. For a driver in the centre of a circular baffle of radius $a$ and a microphone on axis at distance $r$, all edge points have the same path difference

$$\Delta = a + \sqrt{a^2 + r^2} - r \approx a \quad (r \gg a). \tag{1}$$

Because the edge wave is inverted, the two waves are in phase (peak) and out of phase (dip) at

$$f_{\max,n} = (2n-1)\,\frac{c}{2\Delta}, \qquad f_{\min,n} = n\,\frac{c}{\Delta}, \qquad n = 1, 2, \ldots \tag{2}$$

With the driver off-centre on a rectangle (the IEC screen), the edge distances are spread and the secondary waves arrive at many different times, so the ripple averages out. Off axis, the edge points of a circle are no longer equidistant from the microphone, and the ripple is smeared. At low frequency, where the wavelength is large compared with the baffle, the sound wraps round into full space ($4\pi$) and the on-axis level falls by up to 6 dB: the *baffle step*. The course function `diffrac.m` implements Terai's method for a thin baffle with a point source or a piston of radius $a_s$.

### 1.2 Piston directivity

A rigid piston of radius $a$ in an infinite baffle has the far-field directivity

$$D(\theta) = \frac{2J_1(ka\sin\theta)}{ka\sin\theta}, \qquad k = \frac{2\pi f}{c}, \tag{3}$$

which is $-3$ dB at $ka\sin\theta = 1.62$ and has its first null at $ka\sin\theta = 3.83$. Dividing an off-axis response by the on-axis response on the same baffle removes the driver and, to first order, the baffle, and leaves $D(\theta)$.

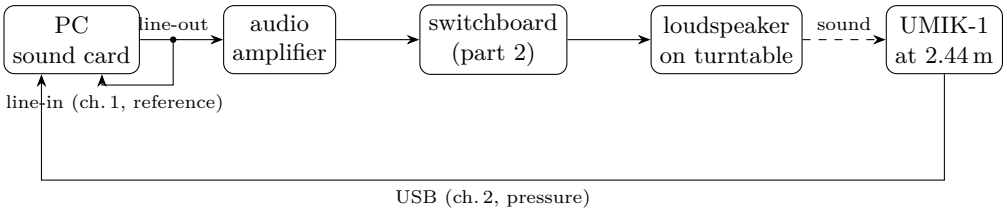
### 1.3 Transfer function and phase

The PC's line-out drives the amplifier and is fed back to line-in (channel 1, the reference); the UMIK-1 is channel 2. The transfer function of unit $k$ in a run is

$$H_k(f) = \frac{\texttt{specn(:,2,k)}}{\texttt{specn(:,1,k)}} \quad \text{[Pa/V]}. \tag{4}$$

Its phase contains the unit's own phase, the flight time $r/c$, and a random delay $\tau_0$ that the sound card adds to every run because playback and recording are not synchronised. Within one run ($N_{\text{meas}} = 3$ through the switchboard) $\tau_0$ and $r$ are common to the three units, so the relative phase $\angle(H_a/H_b)$ is valid. The course function `phaseunwrap.m` adds $2\pi f d/c$ and unwraps; one distance $d$ is used for all units and adjusted until the phases are neither all rising nor all falling.

## 2 Equipment and geometry




| Item | Value |
|---|---|
| Microphone | UMIK-1 (miniDSP) S/N 708-0332, its own correction file, incidence 0ř |
| Settings, part 1 | 50 Hz to 20 000 Hz, 24 lines/octave, $f_{\text{res}} = 1$ Hz, $N_{\text{av}} = 16$, $N_{\text{meas}} = 1$ |
| Settings, part 2 | 50 Hz to 20 000 Hz, 48 lines/octave, $N_{\text{av}} = 64$, $N_{\text{meas}} = 3$ (woofer, midrange, tweeter) |
| Microphone distance | 2.44 m in both parts; height 1.15 m above the platform in part 2 |
| Temperature | 20 °C, so $c = 331.3\sqrt{1 + T/273.15} = 343.2$ m/s |
| 3 " driver | effective diameter 0.06 m ($a \approx 3$ cm, rule of thumb: radius in cm ≈ size in inches) |
| Circular screen | radius $a = 0.30$ m, driver centred |
| IEC screen | 0.51 m × 0.71 m (width × height), driver centre 0.205 m from the left and 0.44 m from the bottom edge |
| System D woofer | Scan-Speak 26W/8534G00, effective diameter 0.211 m (data sheet), 88 L box, vent plugged |
| System D mid/tweeter | DALI Spektor 2: 1 × 5¼ " low/mid driver (effective diameter 0.105 m by the rule of thumb) and 1 × 25 mm soft dome (0.025 m); DALI's own crossover 2600 Hz, 84.5 dB (2.83 V/1 m), 6 Ω |
| Layout | mid/tweeter box on the woofer box at `xrel_mid = 0.12, yrel_mid = 0.29`; dust cover off |

The DALI figures are from the manufacturer's specification page (https://www.dali-speakers.com/en-us/products/discontinued/spektor/spektor-2/). Diameters of the midrange and tweeter were not measured in the lab.

## 3 Measurement record and quality checks

| File (`labE_<tag>.mat`) | What | Note |
|---|---|---|
| `box_0`, `box_30` | 3 " driver in its small box | |
| `circle_0`, `circle_30` | circular screen | |
| `iec_0/15/30/60` | IEC screen, off-centre | |
| `system_0`, `system_30` | System D, three units per file | used |
| `system_30_bad` | System D, 30ř | rejected: up to 31 dB off `system_30` |

- The 60ř system run was not taken.
- Above 18.4 kHz the UMIK channel rises by about 20 dB in every run, including the bare-driver runs, while the reference channel stays flat. This is a measurement artefact, so all results are shown up to 18 kHz.
- The chamber is not anechoic at low frequency, and in part 2 the vent was plugged. Below about 100 Hz the curves describe the room and the mounting rather than the drivers; the tweeter's response below 100 Hz is noise.

- The reference is the line-out *before* the amplifier, so every $|H|$ includes the amplifier gain. The absolute levels in dB re 1 Pa/V can therefore be compared between runs, but not with a manufacturer's sensitivity.

# 4 Part 1: one driver, three mountings

## 4.1 Comparison of the mountings

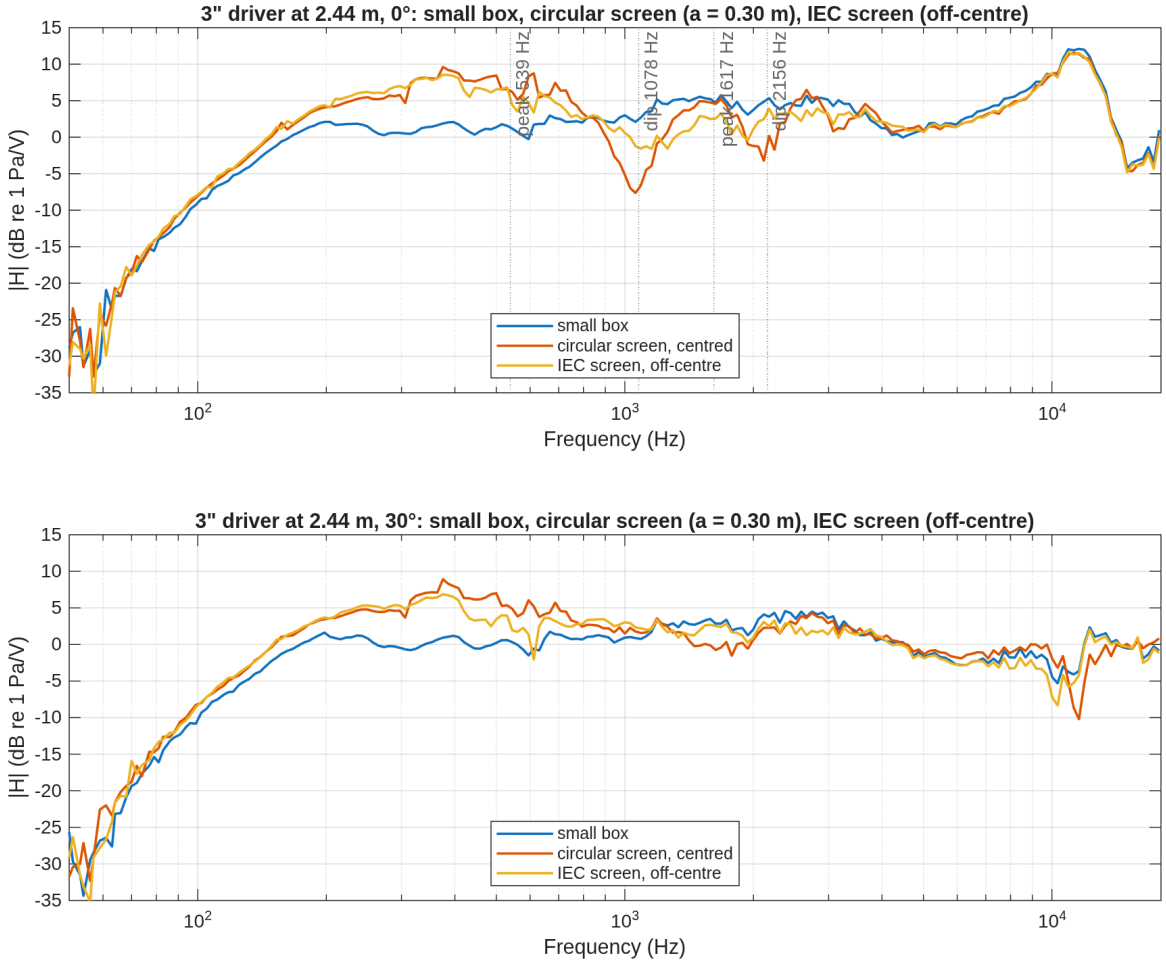



Figure 1: the 3 " driver at 2.44 m on the three mountings, 0ř (top) and 30ř (bottom). Dotted lines: the circle's first two peaks and dips from Eq. (2).

| | small box | circular screen | IEC screen |
|---|---|---|---|
| ripple 300 Hz to 5000 Hz, peak to peak, 0ř | 6.0 dB | 17.3 dB | 10.2 dB |
| ripple 300 Hz to 5000 Hz, peak to peak, 30ř | 6.1 dB | 10.5 dB | 9.0 dB |

- **Small box.** From 250 Hz to 600 Hz it is 5.7 dB below the IEC screen, and from 3 kHz to 8 kHz it is level with it (−0.3 dB). This is the baffle step: the box front is too small to keep the radiation in half space at low frequency. Its edges are close to the driver, so its own ripple is small and at higher frequency.
- **Circular screen.** With all edge points equally far away the secondary waves add up, and the circle has by far the largest ripple, with deep dips (Section 4.2).
- **IEC screen.** The edges are 0.20 m to 0.44 m from the driver, so the ripple is spread and smaller than on the circle, with no deep dip.
- **At 30ř** the circle's ripple falls from 17.3 dB to 10.5 dB, and above 1 kHz the three mountings lie within a few dB of each other.
- Common to all three: the driver's own response, a roll-off below about 200 Hz and a peak near 11 kHz.

## 4.2 Circular screen: theory and diffrac.m

With $a = 0.30\,\text{m}$ and $r = 2.44\,\text{m}$, Eq. (1) gives $\Delta = 0.318\,\text{m}$.

| | Eq. (2) | measured, circle 0ř |
|---|---|---|
| first peak ($\Delta = \lambda/2$) | 539 Hz | broad maximum 400 Hz to 600 Hz |
| first dip ($\Delta = \lambda$) | 1078 Hz | 1059 Hz, −7.6 dB |
| second peak ($\Delta = 3\lambda/2$) | 1617 Hz | 1679 Hz |
| second dip ($\Delta = 2\lambda$) | 2156 Hz | 2115 Hz |

The dips are within 2 % of the formula. The first peak is not sharp, because the driver's own response and the baffle step change across the same octave.

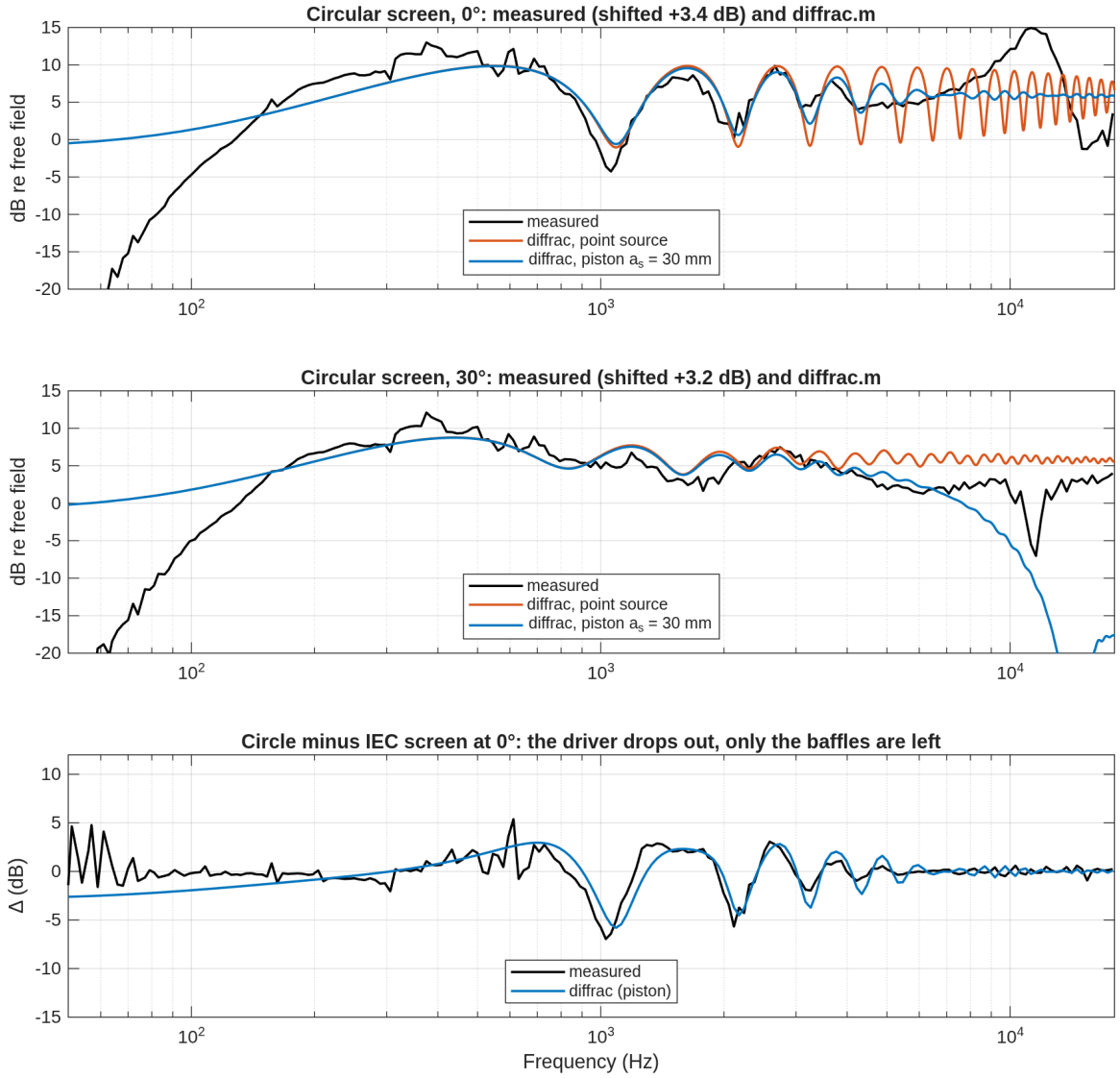



Figure 2: circular screen at 0ř and 30ř against `diffrac.m` for a point source and a piston with $a_s = 30\,\text{mm}$ (top two panels; the measured curve is shifted by 3.4 and 3.2 dB onto the model's mean over 300 Hz to 5000 Hz). Bottom: circle minus IEC screen at 0ř.

- `diffrac.m` returns the level relative to free field, so the infinite-baffle level is 6 dB. The measured curves also contain the driver; they are shifted onto the model, and their shape is compared.
- The point source keeps rippling at full amplitude to 20 kHz. The 30 mm piston radiates less along the baffle once $ka > 1$ ($ka = 1$ at 1.8 kHz), so the edge waves and the ripple weaken at high frequency. The measurement follows the piston.
- **The cleanest comparison** is the bottom panel: the same driver is in both the circle and the IEC measurement, so it cancels in the difference. Measured and modelled circle-minus-IEC agree to 1.4 dB RMS (maximum 3.5 dB) from 300 Hz to 5000 Hz, including the dips at 1029 and 2115 Hz.

- Below 300 Hz the model and measurement part, because of the driver roll-off and the stand, which the thin-baffle model does not include.

## 4.3 IEC screen: directivity

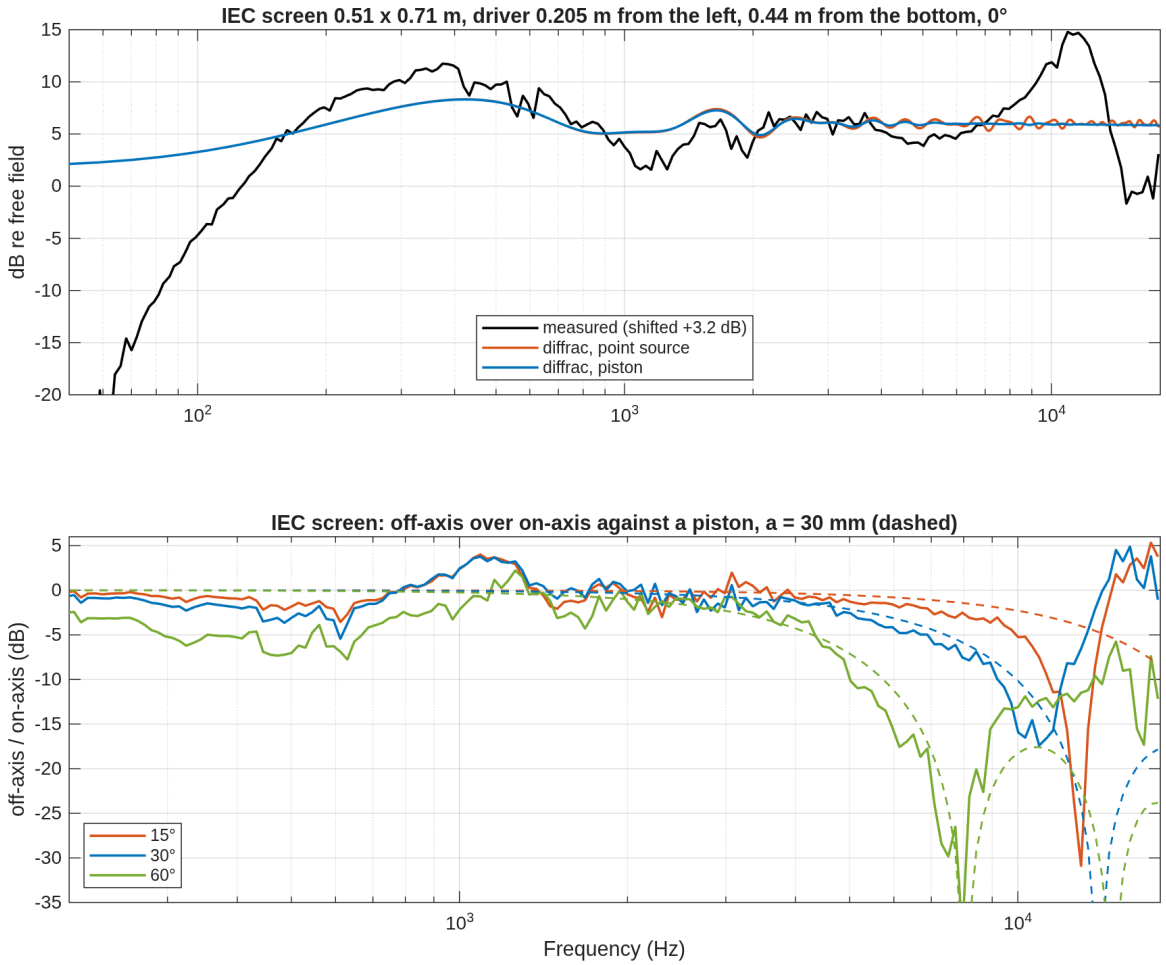



Figure 3: top, IEC screen at 0ř against `diffrac.m` (rectangle defined relative to the driver). Bottom: 15ř, 30ř and 60ř divided by 0ř (solid) against a 30 mm piston (dashed).

| | piston, $a = 30$ mm: $-3$ dB / null | measured: $-3$ dB / null |
|---|---|---|
| 30ř | 5.9 kHz / 14.0 kHz | 5.2 kHz / 10.9 kHz ($-17$ dB) |
| 60ř | 3.4 kHz / 8.1 kHz | 3.45 kHz / 7.96 kHz ($-38$ dB) |

- At 60ř the measurement lies on the rigid piston up to the null.
- At 15ř and 30ř the curves follow the piston to about 5 kHz. Above it the measured nulls come earlier than the piston's (the 30ř null implies $a \approx 38$ mm, and at 15ř there is a dip at 13 kHz where the piston has none). The cone does not move as a rigid piston there (break-up), which also shows as the peak near 11 kHz in Figure 1.
- The 3 dB bump near 1.1 kHz in every ratio is the IEC screen's own on-axis dip, which the off-axis curves do not share: the normalisation removes the baffle only approximately.

# 5 Part 2: System D

## 5.1 Magnitude

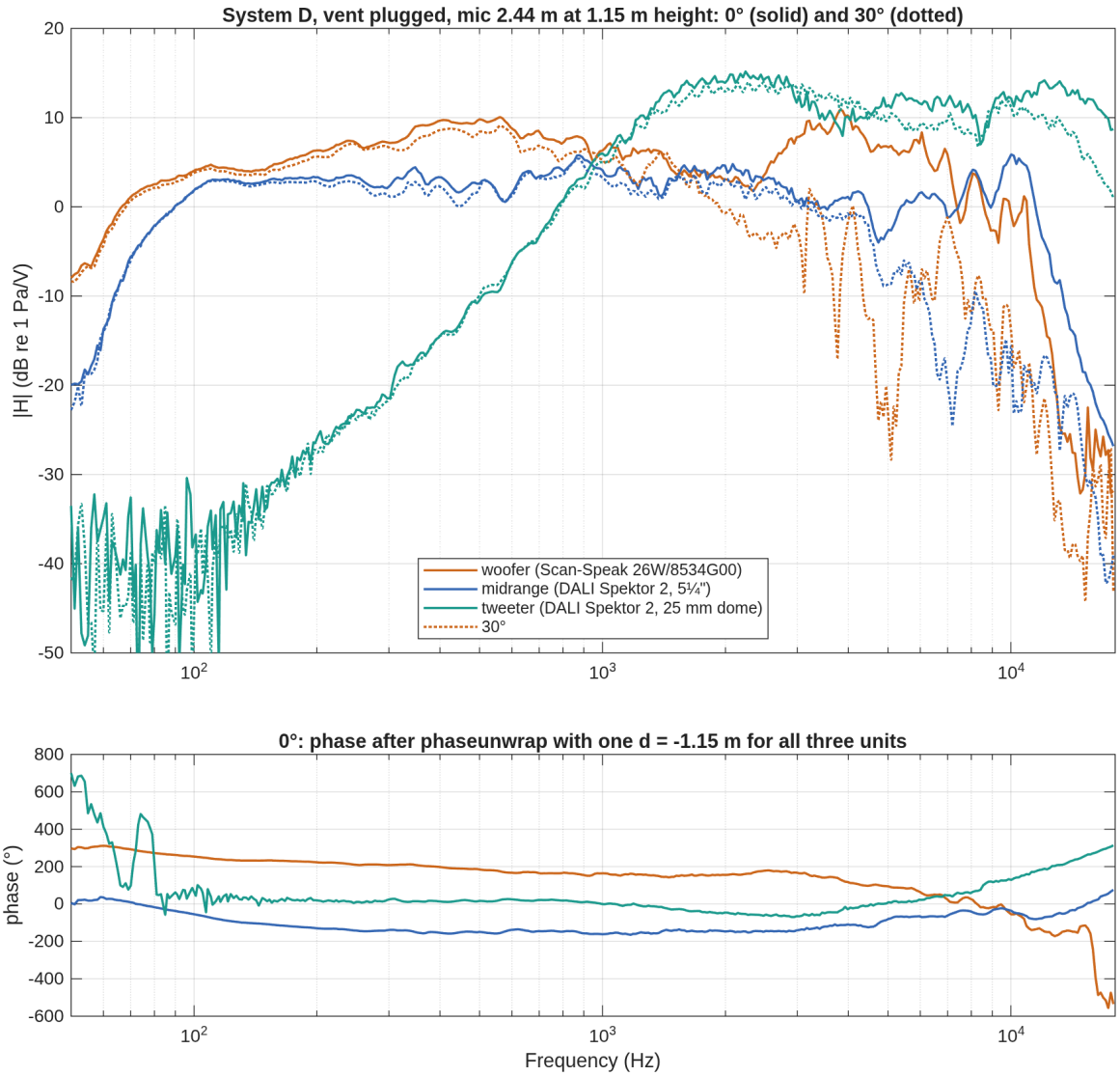



Figure 4: System D, vent plugged. Top: the three units at 0ř (solid) and 30ř (dotted). Bottom: phase at 0ř after `phaseunwrap.m` with one distance $d = -1.15$ m for all units, shifted by whole turns to lie near each other at 1 kHz.

| mean level, dB re 1 Pa/V | 150 Hz to 600 Hz | 1 kHz to 3 kHz | 3 kHz to 8 kHz | 8 kHz to 15 kHz |
|---|---|---|---|---|
| woofer (Scan-Speak 26W) | 7.7 | 4.7 | 6.7 | −9.8 |
| midrange (Spektor 2, 5¼ ") | 2.8 | 3.3 | 0.2 | −2.2 |
| tweeter (Spektor 2, 25 mm) | −19.8 | 12.4 | 11.1 | 12.1 |

- **Woofer**: the most sensitive unit at low frequency, 5 dB above the midrange at 150 Hz to 600 Hz. On axis it extends to about 3 kHz, with break-up peaks at 2.5 kHz to 3 kHz.
- **Midrange**: the flattest unit, within ±2 dB from 120 Hz to 4 kHz, and usable on axis to about 12 kHz.
- **Tweeter**: about 9 dB above the midrange in its band. It is 6 dB down from its band level at 1.04 kHz, just below its resonance (the Lab D impedance has its maximum, 6.7 Ω, at 1.45 kHz).
- **Crossings**: the tweeter crosses the woofer at 958 Hz and the midrange at 944 Hz. A crossover there would drive the tweeter at its resonance. A mid/tweeter crossover around 2.5 kHz to 3 kHz, with about 9 dB attenuation on the tweeter, keeps it an octave clear; DALI's own crossover for the Spektor 2 is at 2600 Hz.

## 5.2 Directivity

| | $a$ | $ka = 1$ at | 30ř minus 0ř, measured (piston) | | |
|---|---|---|---|---|---|
| | | | 1 kHz to 3 kHz | 3 kHz to 8 kHz | 8 kHz to 15 kHz |
| woofer | 105.5 mm | 518 Hz | −3.5 (−4.2) | −15.6 (−23.7) | −12.9 (−31.9) |
| midrange | 52.5 mm | 1.04 kHz | −1.3 (−0.9) | −7.5 (−10.3) | −16.7 (−24.3) |
| tweeter | 12.5 mm | 4.37 kHz | −0.9 (−0.1) | −0.7 (−0.4) | −2.5 (−1.9) |

- The tweeter behaves as a 25 mm piston.
- Woofer and midrange beam as expected up to their first octave above $ka = 1$, but less than a rigid piston above that: once a cone breaks up, only its inner part radiates efficiently, and its effective radius shrinks. The deep piston nulls are also filled by the averaging over each band.
- For the design: the woofer is −3 dB at 30ř near 1.7 kHz to 1.9 kHz and must hand over well below that; the midrange holds up to about 3 kHz, which is consistent with a crossover near 2.6 kHz.

## 5.3 Phase

- **Compensation distance.** The single $d$ that makes the three phases smoothest (minimum summed phase step between adjacent lines, 200 Hz to 15 000 Hz) is $d = -1.15$ m; the midrange and tweeter alone give −1.16 m, the woofer −1.10 m.
- $d$ is negative although the microphone was 2.44 m away, because it absorbs the sound card's random delay as well as the flight time. For the same reason the other runs give unrelated values (`system_30`: −7.3 m; `box_0`: −6.8 m).
- At $d = -1.15$ m the midrange slope is slightly rising and the tweeter slope slightly falling between 1 and 4 kHz, which is the brief's stopping criterion (mixed slopes).

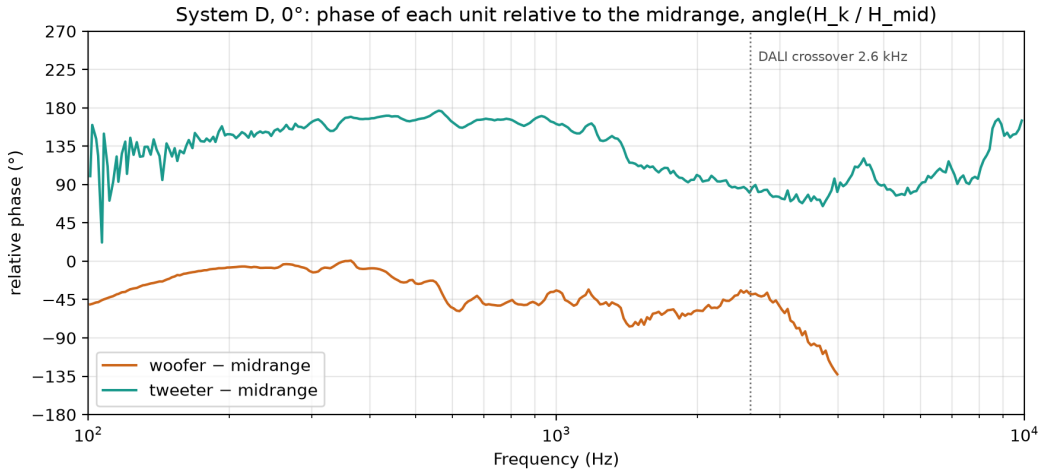



Figure 5: phase of the woofer and the tweeter relative to the midrange at 0ř, $\angle(H_k/H_{\mathrm{mid}})$, from one run. Every common delay cancels, so no compensation distance is needed.

| $f$ | 300 Hz | 1 kHz | 2 kHz | 2.5 kHz | 3 kHz | 4 kHz |
|---|---|---|---|---|---|---|
| woofer − midrange | −12° | −36° | −58° | −37° | −54° | −134° |
| tweeter − midrange | +163° | +162° | +97° | +85° | +76° | +90° |

- **Woofer and midrange** are close to in phase at low frequency, and the woofer lags increasingly with frequency: its acoustic centre lies further back, and its low-pass roll-off adds phase lag.
- **The tweeter** is nearly inverted relative to the midrange below 1 kHz, as expected far below the resonance of a second-order high-pass system (180°), and it comes round to about 85° at 2.6 kHz.
- The crossover filters' own phase shifts and the drivers' polarities therefore have to be chosen against this measured relative phase. A plain in-phase assumption would not sum correctly in the crossover region.

# 6 Connection to Lab D

- Lab D measured the electrical side of the same units: the woofer's free-air resonance $f_S = 22.7\,\text{Hz}$, closed-box resonance $f_C = 42.0\,\text{Hz}$ in the 88 L box, and vent tunings $f_B = 23.4/22.8/21.5\,\text{Hz}$ for 160/200/240 mm tubes. It also gave the complex impedances of the midrange ($R_E = 4.6\,\Omega$, resonance 90 Hz in its box) and the tweeter ($R_E = 3.4\,\Omega$, resonance 1.45 kHz).
- Lab E gives the acoustic side: the magnitude, directivity and relative phase that the crossover must combine.
- The system response is the crossover's voltage transfer (which depends on the Lab D impedances) times each unit's Lab E pressure per volt, summed with their phases.
- Below about 100 Hz, where the chamber is not anechoic and the vent was plugged, the woofer's response has to come from the Lab D tuning and near-field measurement.

# 7 Uncertainty and conclusion

**Main uncertainties.**

- **Geometry.** The mic distance is a single taped value. The 3 " and DALI diameters are rule-of-thumb or data-sheet values rather than measurements. The turntable angle is read by eye ($\pm 2°$).
- **Room and mounting.** The chamber's lower limit and the stand and platform reflections.
- **Measurement chain.** The UMIK correction curve and the artefact above 18.4 kHz; noise where a unit is far below its band (the tweeter under 300 Hz).
- **Phase.** The relative phase depends on the microphone not moving within the run; it does not depend on $d$.

**Conclusion.** A centred driver on a circular baffle shows the regular edge-diffraction pattern predicted by the path difference: dips at 1059 and 2115 Hz against 1078 and 2156 Hz, and 17 dB of ripple. The off-centre IEC screen reduces it to 10 dB, and the small box shows the baffle step (5.7 dB). `diffrac.m` with a 30 mm piston reproduces the baffle difference to 1.4 dB RMS, and the off-axis ratios follow a 30 mm piston (exactly at 60ř) until the cone breaks up. For System D the woofer, the Spektor 2 midrange and the tweeter were measured in one run. The tweeter is 9 dB more sensitive than the midrange and resonates at 1.45 kHz, which points to a mid/tweeter crossover near 2.6 kHz — DALI's own choice. The measured relative phase (tweeter +85° on the midrange at that frequency) is the input the crossover design needs.